Deepak Kharol

Staff Software Engineer · Backend & Distributed Systems
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Summary

Staff engineer with 4+ years building the message-delivery and real-time decisioning core of a customer-engagement platform (Java, MongoDB, Redis) sending billions of messages a month for 2,000+ brands. I own features end-to-end across independently deployed services and build them as extensible, configurable foundations the next feature reuses. My work runs in production for Flipkart, Nykaa, Kotak Mahindra Bank and Axis Bank.

Highlights

- Built the per-user frequency-capping engine that enforces message budgets in memory on the send path; live for 60+ enterprise accounts within six months and the base of four later products.
- Technical lead for a five-service, event-driven price-drop and back-in-stock trigger for Flipkart (500M+ registered users, 150M-product catalog, ~50M catalog updates a day), built for Big Billion Days scale.
- Built the variant-selection and budget-simulation layer of the AI decisioning agents now sold as a paid add-on.

Experience

CleverTap — *Mumbai, India*

Staff Software Engineer — *Jul 2026 – Present*

- **Price-drop / back-in-stock trigger (Flipkart).** Led the design of a pipeline that alerts shoppers when a product they viewed drops in price or restocks, for a 150M-product catalog taking ~50M updates a day. Built the delivery side for millions of qualifying users per sale: streaming reader, bounded concurrency, disk-spilling de-duplication, back-pressure on a shared queue, fail-closed freshness gates. Thresholds, price bands and pacing are per-account configuration changeable mid-sale without a deploy; capture is a pluggable hook any ingestion path can supply. 17 stacked PRs, safe in any deploy order.
- **AI campaign ranking and 1:1 individualisation.** The platform's decisioning layer runs reinforcement-learning agents, each optimising a reward function (clicks, conversions, revenue), to rank every campaign a user qualifies for and send the one most likely to earn that reward. Extended it to choose the content variant per user, replacing one A/B winner for everyone: designed the versioned ranking store (headroom so later fields need no format bump), the contract and a fail-open resolver. Sold as the Decisioning Agents add-on; first deployments with Axis Bank and Nykaa.
- **Message spacing and resilience.** Messages blocked by a per-user time gap are re-queued for the next eligible minute (quiet hours and delivery windows respected) instead of dropped; scheduled campaigns survive event-store outages via configurable backoff retry.

Senior Software Engineer — *Jan 2024 – Jun 2026*

- **Frequency capping.** Designed and built a credit-based engine for rule-driven per-user message budgets. Channel, label, segment and team are optional rule criteria and daily, weekly, monthly and per-slot windows are plug-in limit types, so new rule kinds need no dispatch-path change. Each user's state is a compact byte array rebuilt nightly from 30 days of send history, so a cap check is a few array reads with no database call. Real-time rule propagation to every shard; idempotent legacy migration; budget simulator for the AI campaign ranker. Live for 60+ accounts including Kotak Mahindra Bank group (50M+ customers), Axis Bank (~59M customers), Nykaa and Bajaj Finserv.
- **Campaign data archival.** Designed a reusable collection-archival framework (configurable age threshold, batch size, schedule) moving stale campaigns and statistics from the primary MongoDB cluster to a low-index archive, transparently.
- **Server-side unified inbox.** Owned the backend of a Redis-backed, user-level in-app inbox channel built for StockX (CRUD, dispatch, live triggers, reporting, billing, version history, outage retry); supported a $1M ARR migration.
- **Reliability and security.** Fixed a 4.7 GB heap out-of-memory on the dispatch path by replacing a non-thread-safe LRU cache with a lock-free one; made audit writes asynchronous; completed PII encryption in the send path for regulated clients (Kotak Mahindra Bank group, Axis Bank, Flipkart, EU accounts under GDPR).
- **Leadership.** Point of contact for a 10-engineer team; mentored three engineers who shipped five features in two months; on-call owner.

Software Engineer — *Jul 2022 – Dec 2023*

- **Campaign creation API.** Brought the programmatic API to parity with the dashboard for push, email and SMS as a per-section handler pipeline; now creates 40%+ of all campaigns; v2 caps, validation and compression plugged in later.
- **Send-time content personalisation.** Architected fetching content from client APIs at send time with Liquid templating of URL parameters and JSON responses, so brands personalise without syncing data first.
- **Analytics, integrations, security.** Push render-rate measurement; phone-number matching for Facebook and Google Ads audiences; penetration-test remediation (label and CSV injection).

Backend Intern — built the first campaign API extensions (audience, schedule, setup sections) — *Dec 2021 – Jun 2022*

Dell Technologies, Business Intelligence Intern — Power BI and Tableau dashboards — *May 2021 – Jul 2021*

Technical Skills

Stack	Java 17, Python, C++, SQL · MongoDB (sharded), Redis, RocksDB, PostgreSQL, OpenSearch · Guice, Vert.x, Spring, Docker, AWS, Prometheus/Grafana, Splunk, LaunchDarkly, GitHub Actions
Design	Event-driven pipelines, sharded in-memory state, streaming with back-pressure, at-least-once delivery, versioned formats, flagged rollouts

Education and Awards

KIIT, Bhubaneswar — B.Tech in Computer Science, CGPA 9.36 (2018–2022) · Hackathon winner (2025) · Merit Scholarship